

To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char path[100];

    // Save current directory
    if (getcwd(path, sizeof(path)) == NULL) {
        printf("Error getting directory\n");
        return 1;
    }

    printf("Current directory: %s\n", path);

    char choice[20];

    while (1) {
        printf("\nEnter command (pwd/exit): ");
        scanf("%19s", choice);

        if (strcmp(choice, "pwd") == 0) {
            if (getcwd(path, sizeof(path)) != NULL)
                printf("Current directory: %s\n", path);
        }
    }
}
```

```

        printf("Exiting program...\n");

        // Cleanup before exit
        printf("Resources cleaned up.\n");

        exit(0);
    }
    else {
        printf("Invalid command.\n");
    }
}

return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 5$ ./task3
Current directory: /home/geya/OS/Skill/Week 5

Enter command (pwd/exit): pwd
Current directory: /home/geya/OS/Skill/Week 5

Enter command (pwd/exit): exit
Exiting program...
Resources cleaned up.
geya@LenovoL480:~/OS/Skill/Week 5$ |

```